

Signal Feature Extraction Reference

MIT Short-Circuit Flux and Current Signals — Interturn Fault Dataset (Kaggle)

1. Time-Domain Features — Central Tendency & Dispersion

Table 1: Statistical features measuring signal level and energy content

Feature	Formula	Physical Meaning for Motor Signals
Mean	$\mu = \frac{1}{N} \sum_{i=1}^N x_i$	For AC signals, should be ≈ 0 . Non-zero $\rightarrow$ DC offset or sensor bias. Low discriminative power alone.
RMS (Root Mean Square)	$x_{RMS} = \sqrt{\frac{1}{N} \sum_{i=1}^N x_i^2}$	Most important! Represents energy content. Faults increase current RMS (more current drawn) and alter flux RMS.
Standard Deviation	$\sigma = \sqrt{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^2}$	Measures signal spread. For zero-mean AC signals, $\sigma \approx \text{RMS}$. More useful when a DC offset is present.
Peak-to-Peak	$x_{pp} = \max(x) - \min(x)$	Signal amplitude range. Short-circuit faults cause higher current peaks.

2. Time-Domain Features — Shape Statistics (Higher Moments)

Table 2: Higher-order moments for fault asymmetry and peakedness detection

Feature	Formula	What It Detects
Skewness	$Skew = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^3}{\sigma^3}$	Asymmetry of the distribution. Normal AC signals are symmetric (Skew ≈ 0). Faults introduce asymmetry (e.g., half-wave rectification effect).
Kurtosis	$Kurt = \frac{\frac{1}{N} \sum_{i=1}^N (x_i - \mu)^4}{\sigma^4}$	"Peakedness." Excellent for intermittent faults. Normal: ≈ 3 (Gaussian). HI faults: > 5 (leptokurtic — sharp spikes). LI faults: < 2.5 (platykurtic — clipped).

3. Time-Domain Signal Characteristics

Table 3: Waveform shape descriptors sensitive to distortion and fault severity

Feature	Formula	Application
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Crest Factor	$CF = \frac{\max x }{x_{RMS}}$	Peak-to-RMS ratio. High CF → impulse spikes (HI faults). Low CF → smooth waveform (LI faults).
Form Factor	$FF = \frac{x_{RMS}}{\frac{1}{N} \sum_{i=1}^N x_i }$	Wave shape purity. Pure sine wave: $FF \approx 1.11$. Distorted waveforms deviate from this value.
Clearance Factor	$CLF = \frac{\max x }{\left(\frac{1}{N} \sum_{i=1}^N \sqrt{ x_i }\right)^2}$	Sensitive to early-stage faults. Effective for detecting incipient degradation before RMS changes.
Zero-Crossing Rate	$ZCR = \frac{\# \text{ sign changes}}{N}$	Times the signal crosses zero per sample. Changes with frequency content and harmonic distortion.
Signal Energy	$E = \sum_{i=1}^N x_i^2$	Total energy of the signal window. Increases significantly with fault severity.

4. Frequency-Domain (Spectral) Features

Table 4: Spectral features derived from the FFT of the signal

Feature	Formula	Description	Fault Indication
Fundamental Frequency	$f_1 = \operatorname{argmax} X(f) , \quad f \in$	Dominant frequency (30–60 Hz based on driver)	Should match set frequency. Deviation indicates a problem.
Harmonic Magnitudes	$ X(h \cdot f_1) , \quad h = 2, 3, \dots$	Amplitude at harmonic multiples	Short-circuits significantly increase odd harmonics.
Total Harmonic Distortion	$THD = \frac{\sqrt{\sum_{h=2}^H V_h^2}}{V_1}$	Ratio of harmonic to fundamental power	Higher THD = more distortion = fault presence.
Spectral Centroid	$f_c = \frac{\sum_{\square} f \cdot X(f) }{\sum_{\square} X(f) }$	"Center of mass" of the spectrum	Shifts to higher frequencies with fault severity.

Spectral Spread	$f_{ss} = \sqrt{\frac{\sum_{\square} (f - f_c)^2 \cdot X(f) }{\sum_{\square} X(f) }}$	Spectrum width around centroid	Increases with added harmonics and noise.
Sideband Ratio	$SR = \frac{A_{\text{left}} + A_{\text{right}}}{2 \cdot A_{\text{fund}}}$	Sideband amplitude vs. fundamental	Normal: < 0.05 Fault: > 0.15

5. Three-Phase Specific Features (Current & Flux Signals)

Table 5: Features leveraging the three-phase nature of the motor signals

Feature	Formula	Physical Meaning
Current RMS per Phase	$I_{RMS, a} = \sqrt{\frac{1}{N} \sum_{i=1}^N i_a^2}$	RMS value computed independently for phases A, B, and C.
Phase Imbalance	$Imb = \frac{\max(I_{RMS}) - \min(I_{RMS})}{\text{mean}(I_{RMS})}$	Key fault indicator. Normal: < 5%. Fault: > 15%. Interturn faults break magnetic symmetry.
Negative Sequence Current	$I_2 = \frac{I_a + a^2 I_b + a I_c}{3}, \quad a = e^{j\frac{2\pi}{3}}$	Classic fault indicator. Normal: near 0. Fault: > 5% of positive sequence. Directly reflects asymmetry from shorted turns.
Park's Vector Magnitude	$PV = \sqrt{i_a^2 + i_\beta^2}$	Circular pattern in healthy operation; becomes elliptical with interturn faults (Clarke transform of i_a, i_b, i_c).
Flux–Current Phase Angle	$\theta_{\varphi, i} = \angle \varphi - \angle i_a$	Phase lag between leakage flux φ and phase current i_a . Changes measurably with turn count — complement to correlation ρ .

6. Wavelet-Domain Features

Table 6: Wavelet decomposition captures non-stationary transient fault events missed by FFT

Feature	Formula	Why It Matters for Interturn Faults
Wavelet Sub-band Energy	$E_j = \sum_k d_{j,k} ^2$	Interturn faults produce transient spikes localised in time. Sub-band energy (e.g., db4, levels 4–6) captures energy where FFT cannot:

		simultaneously in time and frequency.
Wavelet Energy Ratio	$WER_j = \frac{E_j}{\sum_j E_j}$	Relative contribution of each sub-band to total signal energy. Fault severity shifts energy from low to high sub-bands — a strong discriminative feature.
Wavelet Entropy	$WE = - \sum_j p_j \log_2(p_j)$	Measures disorder in the wavelet energy distribution. Healthy signal has low entropy (energy concentrated in fundamental). Faults spread energy → higher entropy.

7. Instantaneous Features (Hilbert Transform)

Table 7: Useful when drive frequency varies or when tracking envelope modulation

Feature	Formula	Application
Instantaneous Frequency	$f_{inst}(t) = \frac{1}{2\pi} \cdot \frac{d\varphi(t)}{dt}$	Tracks real-time frequency deviation. Useful if the drive speed varies during measurement. Fault-induced torque ripple modulates instantaneous frequency.

Notation Reference

- N = number of samples in window | x_i = signal sample at index i | $X(f)$ = FFT of $x(t)$
- μ = signal mean | σ = standard deviation | φ = leakage flux signal
- $i\alpha, i\beta$ = Clarke ($\alpha\beta$) transform components of three-phase currents
- $d_{j,k}$ = wavelet detail coefficients at scale j , position k | p_j = normalised wavelet energy at scale j
- $a = e^{(j2\pi/3)} = e^{(j120^\circ)}$ — symmetrical component rotation operator
- ★ = features added beyond the original set based on interturn fault characteristics